



**The University of Azad Jammu and Kashmir,
Muzaffarabad**

Lab Title: Basic Router Access and CLI Configuration
Course Name: Database Systems
Lab No: 06
Course Code: CS-2203
Submitted to: Engr. Dr. Asma Javaid

Student Name: Kamal Ali Akmal
Roll No: 2024-SE-38
Session: 2024-2028
Semester: 4th
Submission date: 7th May 2026

Department of Software Engineering

Basic Router Access and CLI Configuration

Introduction

Network devices such as routers and switches must be configured before they can operate in a network. Configuration is usually done through the Command Line Interface (CLI). There are two main ways to interact with networking devices:

- GUI (Graphical User Interface)
- CLI (Command Line Interface)

In networking, CLI is widely used because it provides more control and flexibility.

Connecting to Network Devices

To configure routers or switches directly, a computer must be connected to the device using a console connection.

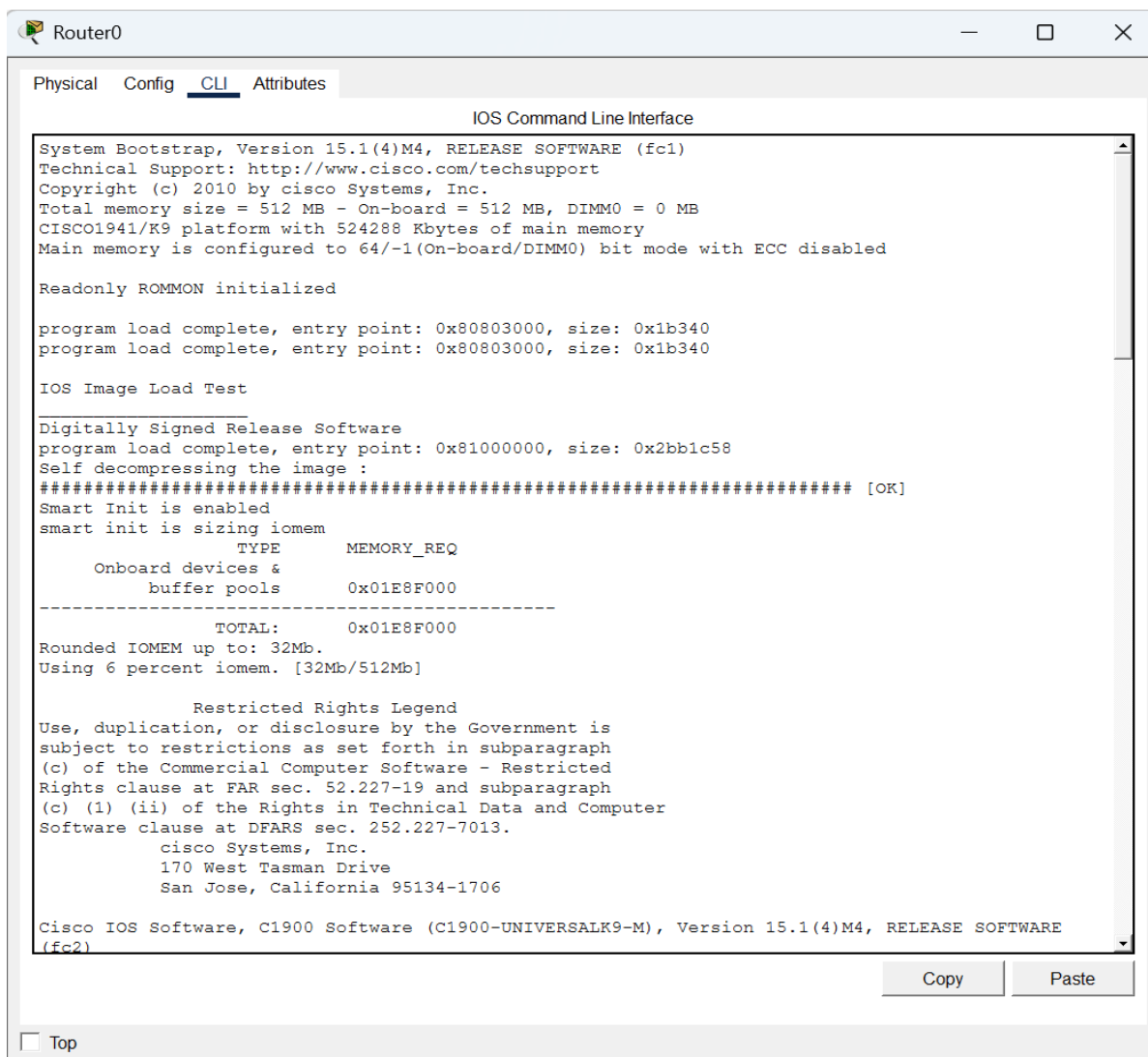
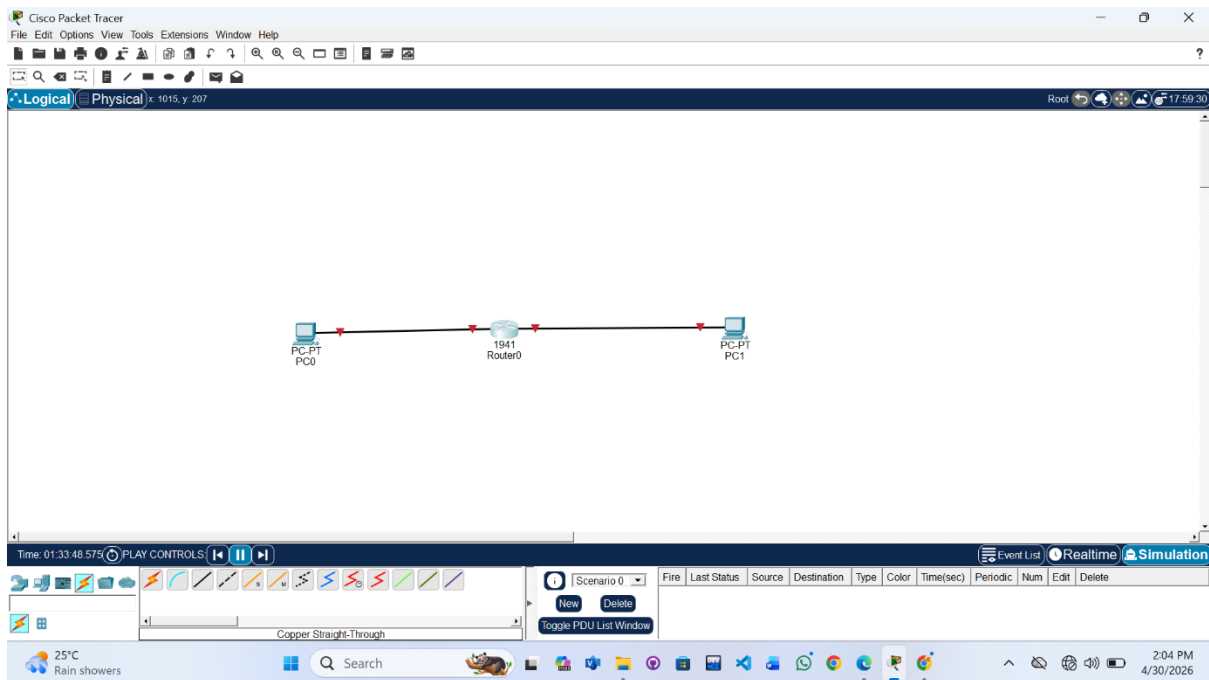
Common connection components:

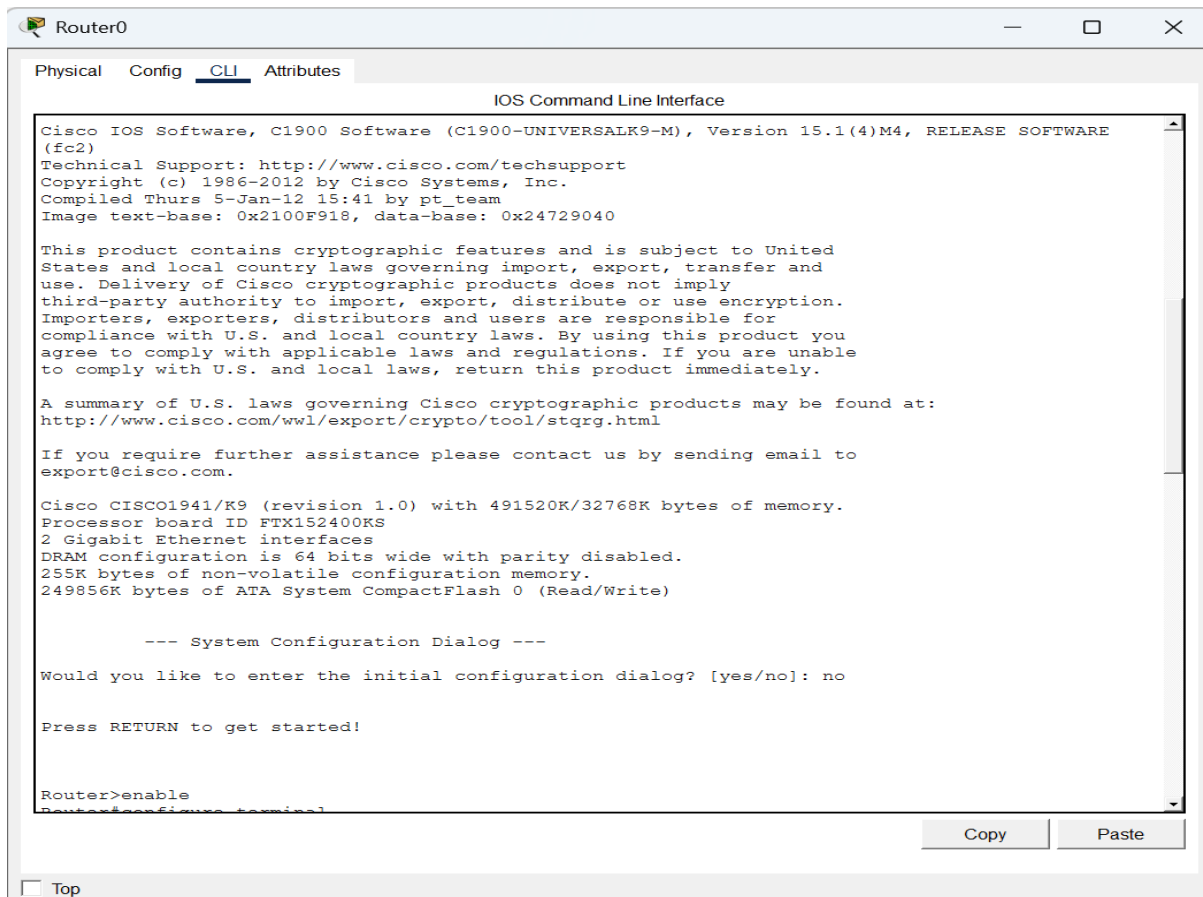
- Console Port on router/switch
- RJ45 Console Cable
- DB9 Connector (older PCs)
- USB Mini Console Port (newer devices)
- Rollover Cable used for console connection

Task

Step 1: Open Router

- Go to Cisco Packet Tracer
- Drag a router
- Click it → go to CLI tab





Step 2: Enter Modes, Set passwords and Encrypt passwords

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)#
```

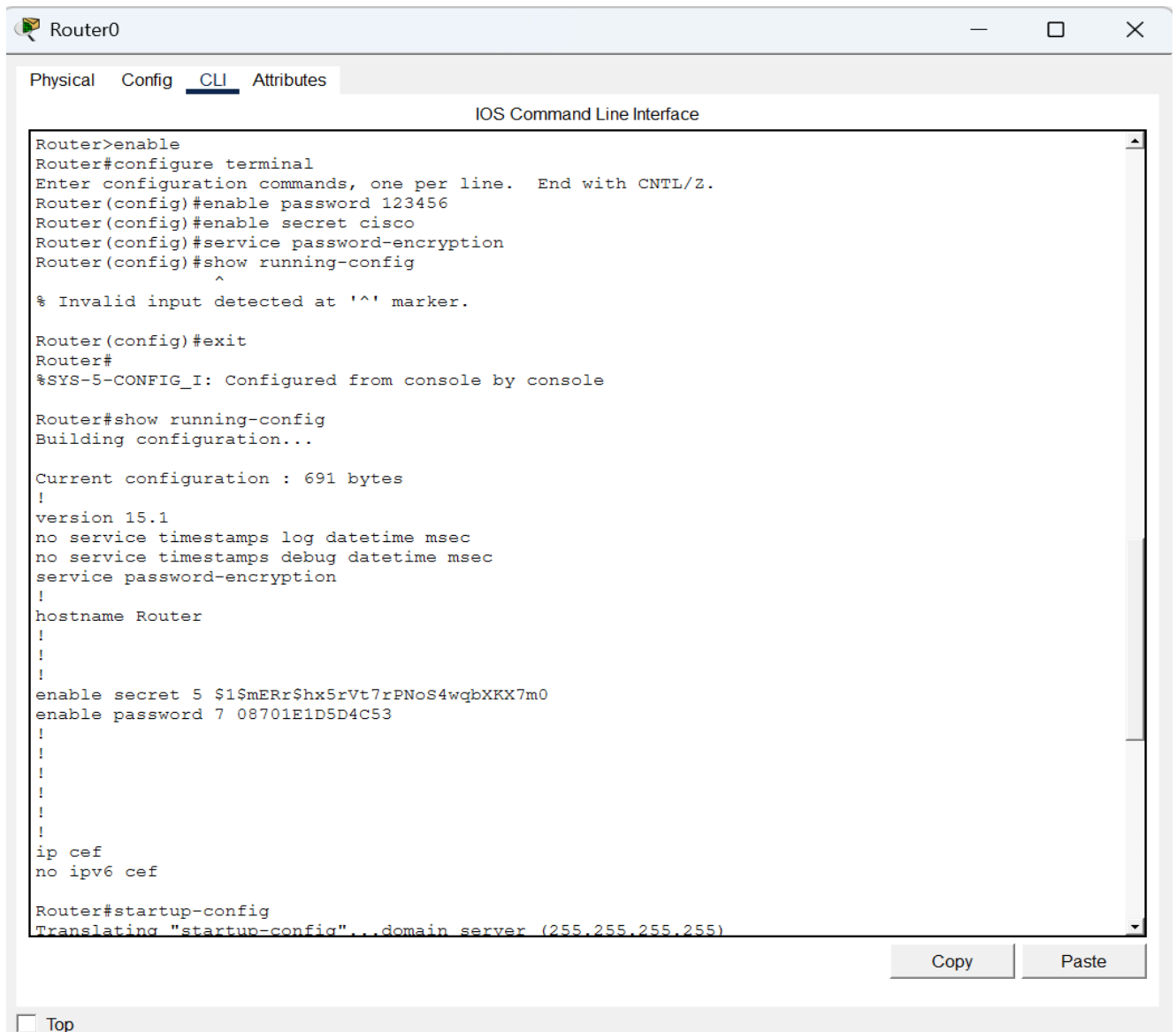
```
Router(config)# enable password 123
```

```
Router(config)# enable secret 456
```

If both exist → enable secret overrides password

```
Router(config)# service password-encryption
```

This converts visible passwords into encrypted form



The screenshot shows a Cisco Router CLI window titled "Router0". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The main area is titled "IOS Command Line Interface". The terminal output shows the following sequence of commands and responses:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable password 123456
Router(config)#enable secret cisco
Router(config)#service password-encryption
Router(config)#show running-config
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show running-config
Building configuration...

Current configuration : 691 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
enable password 7 08701E1D5D4C53
!
!
!
!
!
ip cef
no ipv6 cef

Router#startup-config
Translating "startup-config"...domain server (255.255.255.255)
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. At the bottom left of the window, there is a "Top" button.

Step 3: Check configurations

```
Router# show running-config
```

```
Router# show startup-config
```

